

Technical Requirements: Modular Micro-Hotel Pod System

1. General Structural & Compliance Standards

- **Fire Protection:** All units must meet a minimum of **F30 (fire-retardant)** classification. The shell must act as a self-contained fire compartment.
- **Acoustic Insulation:** * **Airborne Noise:** High-performance sound reduction (target 40 DB) to isolate urban noise and neighboring pods.
 - **Structure-borne Noise (Körperschall):** Total vibration isolation from the host building floor (e.g., via elastomeric bearings).
- **Infrastructure Impact: "Plug & Play"** design or modular assembling on-site. No destructive intervention to the host building's floor, walls, or ceiling.
- **Legal Classification:** Engineered to comply with German *Beherbergungsstättenverordnung* (Accommodation Regulations) or as "movable assets/furniture" depending on the regulations of the province / Bundesland.

2. Module Specifications

Sleeping Unit

- **Dimensions:** Variable width (90cm to 140cm) for single/double occupancy.
- **Climate Control:** Integrated decentralized AirCon (HVAC) in every unit.
 - *Requirement:* Must offer independent heating/cooling.
 - *Ventilation:* Air exchange system must be configurable—either tapping into central building HVAC or a parallel system exchanging air with the host room.
- **Materials:** Durable, anti-static, dust-resistant, and "hospitality-grade" easy-to-clean surfaces.
- **Fenestration:** Integrated non-opening window to allow natural light while maintaining the fire/acoustic seal.
- **Interior:** Focus on "Cocooning" ergonomics (integrated lighting, charging, storage).

3. Assembly & Logistics

- **Modular In-Situ Assembly:** Units must be deliverable as a **"Kit-of-Parts"**.
 - All components (floor, walls, ceiling) must fit through standard internal door openings.
 - Assembly must be possible on-site without heavy machinery or cranes.
- **Relocation Capability:** Designed for short-term lease cycles (2–5 years). The system must be demountable and transportable to new locations with minimal loss of material.

4. Technology & Automation (The "Mobile-First" Stack)

- **Access Control:** Hardware must support digital door opening (BLE/NFC) via smartphone.
- **Unmanned Integration:** * Built-in CCTV mounts for advanced security.

- Integration with PMS (e.g., Apaleo) for automated guest data verification and check-in.
- **Fire Safety:** Integrated smoke detection and alarm systems networked to a central hub (BMA connection capability).